

Dopant- and defect-related small polarons

Small polarons' associations with dopants and defects are often more complex than those between carriers and dopants in conventional semiconductors. To illustrate these associations, three examples of dopant- and defect-related small polarons are succinctly described. In the simplest situation a small polaron occupies one of the equivalent sites adjacent to a dopant. A more complex case envisions a large polaron being induced to collapse into a small polaron when it is confined to the vicinity of a dopant. The final scenario envisions one of two electrons bound to a defect collapsing to a small polaron as the other electron is liberated.

Dopants in semiconductors whose carriers form small polarons can be more intricately involved in transport than are dopants in conventional high-mobility semiconductors. In particular, electronic transport in conventional high-mobility semiconductors is governed by very low concentrations of carriers and dopants. By contrast, transport in small-polaron semiconductors typically involves very high concentrations of their low-mobility carriers and the dopants which spawn them. As a result direct hops between dopant-related states can dominate transport even well above room temperature. To illustrate these situations an example in which such impurity conduction might prevail is discussed.

5.1 Small polarons bound to dopants and defects

Hole-type charge carriers are introduced in crystalline MnO by replacing a small concentration of Mn^{2+} cations with Li^{1+} cations. Measurements of the dc charge transport (conductivity, Seebeck coefficient, and Hall mobility) indicate that liberated hole-like carriers form small polarons and move by thermally assisted hopping between Mn sites (Crevecoeur and Wit, 1970). Since each liberated carrier is severely localized as a small polaron, a carrier bound to the Li dopant is similarly localized. In particular, a carrier bound to a Li dopant forms a Mn^{3+} cation at one of

the 12 nearly equivalent next-nearest neighbor Mn sites surrounding the Li cation (Bosman and van Daal, 1970). In other words, the dopant state is just a small-polaron bound to a dopant. By contrast, a conventional dopant state, such as the donor state formed by substituting a phosphorus atom for one of silicon's atoms, encompasses very many atoms.

A mixed situation has been reported for holes that are introduced in NiO by substituting Li for Ni (Bosman and van Daal, 1970). DC electronic transport measurements have been interpreted as indicating that liberated carriers move as large polarons. By contrast, these authors envision a hole-like carrier bound to a Li dopant forming a small polaron which resides on one of the 12 Ni next-nearest-neighbor sites that surround the dopant. Hopping of a bound small polaron about its Li dopant contributes to the doped materials' ac conductivity. The large-polaron holes in NiO are ascribed to the wide oxygen *p*-band produced by large ($\sim 1.4 \text{ \AA}$) strongly overlapping oxygen anions. By contrast, NiO's small-polaron holes are attributed to the relatively narrow nickel *d*-bands spawned by weak overlap between small ($\sim 0.7 \text{ \AA}$) nickel cations. Only when a hole is adjacent to a Li-dopant is it stabilized as a nickel-based small-polaron rather than as an oxygen-based large-polaron.

Removing an atom from a material, i.e. replacing the atom with a vacancy, can also introduce charge carriers. In particular, oxygen vacancies are common defects in oxides. In these ionic materials oxygen ions are generally assigned a valence of $2-$. However, an isolated oxygen atom can only bind a single additional electron. The positive charges of adjacent cations must assist in binding a second electron near an oxygen atom to produce its formal valence of $2-$. For example, each oxygen site in SrTiO_3 has two Ti^{4+} nearest neighbors and four Sr^{2+} next-nearest neighbors. In these instances, electrons that are attracted to an oxygen atom and its adjacent cations may remain attracted, albeit more weakly, when the oxygen atom is removed. Thus, an oxygen atom and the corresponding vacancy may both formally bind two electrons.

In principle, two electrons can be liberated from a doubly occupied oxygen vacancy. Moreover, there are two complementary scenarios for thermally induced release of a singlet pair of electrons bound to an oxygen vacancy. In the first circumstance, as with bipolaron formation (discussed in Chapter 7), the energy with which the electron-phonon interaction indirectly binds the vacancy's two electrons together overwhelms their mutual Coulomb repulsion. The energy needed to liberate the first electron is then greater than that to free the second electron. Thus thermal de-trapping of one electron of the pair portends relatively rapid thermal release of the second electron. In the second scenario the two electrons' mutual Coulomb repulsion energy predominates. It then takes much less energy to liberate the first electron than to free the second electron. Freeing just one electron of a

singlet pair localized at an oxygen vacancy not only produces the liberated carrier but concurrently generates a localized spin and optical absorption center associated with the residual bound electron. That is, as one of the pair of localized electrons is freed the remaining electron collapses into a bound small polaron.

Much larger densities of dopants and defects are usually required to significantly alter the electrical conductivity of a small-polaron semiconductor than that of a conventional semiconductor. In particular, a relatively high density of low-mobility small polarons is needed to produce electrical conductivity comparable to that generated by a low density of a conventional semiconductor's high-mobility carriers. The electrical conductivity of a small-polaron material is then insensitive to changes of its carrier density that would greatly affect a conventional semiconductor. Alternatively stated, a small-polaron material's Fermi energy appears "pinned".

5.2 Small-polaronic impurity conduction

Dopants provide charge carriers that can be liberated to move through a host material. Alternatively, *impurity conduction* envisions charge carriers moving directly between dopants. In conventional lightly doped semiconductors (e.g. phosphorus-doped silicon), freed carriers move through the host material with a much greater mobility than that for inter-dopant motion. Thus, impurity conduction only prevails when the temperature is low enough to preclude carriers being freed from their dopants. For example, impurity conduction in conventional semiconductors is only seen at cryogenic temperatures.

A different situation often occurs when charge carriers form small polarons. In these situations the small-polaron material's dopant density is usually orders of magnitude greater than that of common semiconductors. Then dopants are close to one another and charge carriers are never far from dopants. In the terminology of conventional semiconductors, one might say that impurity conduction then dominates charge transport.

To illustrate such situations consider small polarons in heavily doped LaMnO_3 . The nominal (distortion free) LaMnO_3 crystal structure has Mn^{3+} ions at the eight corners of the cubic unit cell, O^{2-} ions centered at the middle of each of the cube's 12 edges, and a La^{3+} ion at the cube's center. Replacing a La^{3+} ion with a divalent ion (Ca^{2+} , Sr^{2+} or Ba^{2+}) introduces a hole. Assuming a random replacement of divalent cations for La^{3+} , Fig. 5.1 shows the probability that a dopant has n of the six nearest La cations also being replaced with dopants. The $n = 0$ curve shows that the probability of finding a dopant without another dopant at a nearest-neighbor dopant position falls rapidly with increasing doping, x . Much attention has focused on $0.2 < x < 0.5$, where this material becomes ferromagnetic. Even at

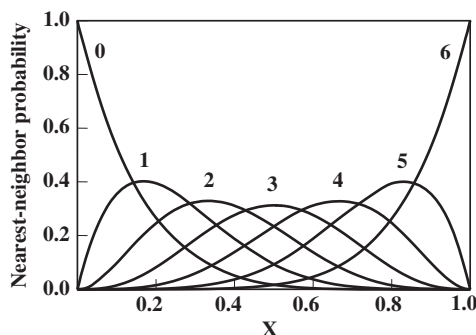


Fig. 5.1 The probability that random substitutions of dopants for La in LaMnO_3 leaves a dopant with n dopant neighbors is plotted against the doping fraction x for $0 \leq n \leq 6$.

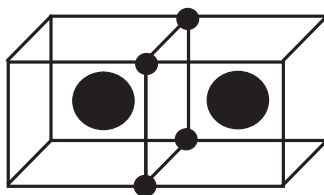


Fig. 5.2 Four Mn sites (small black circles) benefit most from the attractive Coulombic potentials provided by dopants (large black circles) in the centers of two adjacent unit cells of cubic LaMnO_3 .

the lowest doping level in this range 75% of dopants have one or more dopants at nearest-neighbor La sites. That is, dopants are not widely dispersed.

The commonly envisioned small polaron in LaMnO_3 is a hole on a Mn site. The probability of a Mn site not having a dopant occupying at least one of the eight closest La sites is $(1 - x)^8$ (Miller *et al.*, 1961). With $x > 0.2$ this probability is very small, $< (0.8)^8 \approx 0.2$. Thus, a small polaron on a Mn site is generally close to dopants to which it is attracted.

When located between neighboring dopants a Mn site enjoys the sum of their attractive potentials. As illustrated in Fig. 5.2, the potential energy at a Mn cation will be especially low at the four sites that lie between two nearest-neighbor La-site dopants. In fact a magnetic susceptibility study of paramagnetic heavily doped LaMnO_3 concludes that its holes each occupy a localized molecular orbital comprising four Mn cations that align as a ferromagnetic to give a net spin of $4 \times 2 - 1/2 = 15/2$ (Tanaka *et al.*, 1983a). Figure 5.3 depicts the energies of such a cluster's four molecular orbitals. Other locations, such as the two Mn sites comprising the edge shared by three cubic unit cells inhabited by dopants, also

$$2t \text{ — (1)}$$

$$0 \text{ — (2)}$$

$$-2t \text{ — (1)}$$

Fig. 5.3 The molecular energy levels of four Mn cations arranged in a square are split by their nearest-neighbor electronic transfer energy t . The bracketed numbers adjacent to the energy levels indicate the molecular orbitals' degeneracy.

have low potential energies. Holes presumably occupy such low-energy states with multiple holes occupying some clusters of contiguous dopants. In this way carriers in their ground-state distribute themselves in response to the disordered environment generated by the dopants.

Transport measurements in these materials' paramagnetic regime indicate thermally assisted multi-phonon hopping. In particular, analyses of electrical conductivity and Seebeck-coefficient measurements imply that the mobility is low and thermally activated (Heikes and Ure, 1961; Tanaka *et al.*, 1982; Palstra *et al.*, 1997). To obtain the mobility the thermally activated conductivity is presumed to be expressible as $\sigma = nqu$, the product of the carriers' density n , charge q and mobility μ . As explained in Section 12.2, the Seebeck coefficient is approximated as $\alpha = (\kappa/q)\ln[(N - n)/n]$, where N is the density of thermally accessible sites. A thermally activated mobility is inferred by comparing the measured temperature dependence of the conductivity with that of n deduced from the measured Seebeck coefficient.

Seebeck coefficients of heavily doped LaMnO_3 suggest that the phonon-assisted hopping of its holes occurs between a relatively small number of defect-related states rather than among the totality of Mn cations. The magnitudes of the high-temperature Seebeck coefficients are often much too small to just represent the doping concentration divided by the total concentration of Mn sites (Hundley and Neumeier, 1997). For example, the measured room-temperature values of the Seebeck coefficient are $\alpha \sim 5 \mu\text{V/K}$ at 20% doping (Volger, 1954; Tanaka *et al.*, 1983b) while the Seebeck formula gives $\alpha = 120 \mu\text{V/K}$ with $n/N = 0.2$. Moreover, the Seebeck coefficients and electrical conductivities are very sensitive to the introduction of modest densities of oxygen vacancies (Tanaka *et al.*, 1982; Aselage *et al.*, 2003). Transport of Mn-related compensated holes then appears to involve only a subset of Mn sites, e.g. those between adjacent dopants.

The magnitude and temperature dependence of the Hall mobility also indicate phonon-assisted hopping. Early measurements of heavily doped LaMnO_3 also revealed an anomalous signed (n -type for p -type doping) Hall mobility that was

too small ($\ll 1 \text{ cm}^2/\text{V}\cdot\text{sec}$) for conventional transport (Volger, 1954). Subsequent Hall mobility measurements found it anomalously signed, very low and thermally activated with its activation energy well below that deduced for the drift mobility (Jaime *et al.*, 1997). The magnitude and temperature dependence of the Hall mobility is consistent with that for adiabatic small-polaron hopping (Emin and Holstein, 1969). Indeed, the low-mobility thermally activated charge transport deduced for heavily doped LaMnO_3 has often been attributed to small-polaron hopping between transition-metal ions (Volger, 1954; Miller *et al.*, 1961; Kertesz *et al.*, 1982; Tanaka *et al.*, 1982; Palstra *et al.*, 1997; Jaime *et al.*, 1997).

The *n*-type Hall effect observed in heavily doped *p*-type LaMnO_3 suggests hopping among a disordered arrangement of dopant-related localized states. Nearest-neighbor oxygen-mediated small-polaron hopping between Mn sites in LaMnO_3 would occur along its unit-cell's cube edges. However, no Hall-effect sign anomaly should occur for such hopping (Emin, 1971c). In fact, no Hall-effect sign anomaly is found in lightly doped cubic MnO where oxygen-mediated small-polaron hopping also presumably occurs between Mn sites along unit-cell cube edges (Crevecoeur and de Wit, 1970). Rather, Hall-effect sign anomalies are predicted for hopping that is not constrained to be along a cube's edges (Emin, 1977a). Indeed, Hall-effect sign anomalies are frequently observed for hopping conduction in non-crystalline solids.

In summary, electronic transport in the paramagnetic regime of heavily doped LaMnO_3 proceeds via multi-phonon hopping. Although uncertain, the localized states between which carriers hop seem to be dopant-related. An example of such a state is the four-Mn square unit subtended by a pair of dopants depicted in Fig. 5.2. Electronic transport would then constitute a type of small-polaronic impurity conduction.